Solution Sheet 4— Diffusion Processes

4.1: Dynkin's Lemma tells us that since $f \in C_0^2(B_R)$, and $\mathbb{E}\left[\tau_R \wedge t\right] < \infty$ for any t,

$$\mathbb{E}^{\mathbf{a}}\left[f(\mathbf{W}_{\tau_R \wedge t})\right] = f(\mathbf{a}) + \mathbb{E}^{\mathbf{a}}\left[\int_0^{\tau_R \wedge t} \mathcal{A}f(X_s)\mathrm{d}s\right].$$

Since $\mathbf{W}$ is a mulitdimensional Brownian motion we know that $\mathcal{A}f(\mathbf{x}) = \frac{1}{2}\Delta f(\mathbf{x}) = 1$, and so

$$\mathbb{E}^{\mathbf{a}}\left[f(\mathbf{W}_{\tau_R \wedge t})\right] = f(\mathbf{a}) + \mathbb{E}^{\mathbf{a}}\left[\tau_R \wedge t\right].$$

We now note that since $\tau_R \wedge t$ increases with t, we can use the monotone convergence theorem to show that

$$\lim_{t\to\infty} \mathbb{E}^{\mathbf{a}}\left[\tau_R \wedge t\right] = \mathbb{E}^{\mathbf{a}}\left[\lim_{t\to\infty} \tau_R \wedge t\right] = \mathbb{E}^{\mathbf{a}}\left[\tau_R\right].$$

Similarly, since f is bounded in B_R, the bounded convergence theorem gives that

$$\lim_{t\to\infty} \mathbb{E}^{\mathbf{a}}\left[f(\mathbf{W}_{\tau_R \wedge t})\right] = \mathbb{E}^{\mathbf{a}}\left[f(\mathbf{W}_{\tau_R})\right] = \mathbb{E}^{\mathbf{a}}\left[R^2\right] = R^2,$$

by continuity of Brownian motion. We then let $t \to \infty$ in the above and use these results to see that

$$\mathbb{E}^{\mathbf{a}}\left[\tau_R\right] = R^2 - |\mathbf{a}|^2,$$

for any $\mathbf{a} \in B_R$.

4.2: We wish to use Dynkin's Lemma and since we are working with Brownian motion, we know that $\mathcal{A}f(\mathbf{x}) = \frac{1}{2}\Delta f(\mathbf{x})$, so it is simple to show that for $d = 2$ we have

$$\begin{aligned}
\mathcal{A}f(\mathbf{x}) &= \frac{1}{2}\left(\frac{\partial^2 f}{\partial x_1^2}(\mathbf{x}) + \frac{\partial^2 f}{\partial x_2^2}(\mathbf{x})\right) \\
&= \frac{1}{2}\left(\frac{2}{x_1^2 + x_2^2} - \frac{2(x_1^2 + x_2^2)}{(x_1^2 + x_2^2)^2}\right) \\
&= 0.
\end{aligned}$$

Then Dynkin's Lemma tells us that $\mathbb{E}^{\mathbf{x}}\left[f(\mathbf{W}_\tau)\right] = f(\mathbf{x})$.

Similarly, for $d \geq 3$ we have

$$\begin{aligned}
\mathcal{A}f(\mathbf{x}) &= \frac{1}{2}\sum_{j=1}^{d} \frac{\partial^2 f}{\partial x_j^2}(\mathbf{x}) \\
&= \frac{1}{2}\sum_{j=1}^{d}\left(-\frac{d}{2}2(2-d)\left(\sum_i x_i^2\right)^{-\frac{d+2}{2}} x_j^2 + (2-d)\left(\sum_i x_i^2\right)^{-\frac{d}{2}}\right) \\
&= \frac{1}{2}\left(-d(2-d)\left(\sum_i x_i^2\right)^{-\frac{d}{2}} + d(2-d)\left(\sum_i x_i^2\right)^{-\frac{d}{2}}\right) \\
&= 0,
\end{aligned}$$

and then again $\mathbb{E}^{\mathbf{x}}\left[f(\mathbf{W}_\tau)\right] = f(\mathbf{x})$.

We also know that $|\mathbf{W}_\tau| \in \{a, b\}$ almost surely, and so

$$\begin{aligned}\mathbb{E}^{\mathbf{x}}\left[f(\mathbf{W}_\tau)\right] &= f(\mathbf{a})\mathbb{P}^{\mathbf{x}}\left(|\mathbf{W}_\tau| = a\right) + f(\mathbf{b})\mathbb{P}^{\mathbf{x}}\left(|\mathbf{W}_\tau| = b\right) \\ &= f(\mathbf{a})\mathbb{P}^{\mathbf{x}}\left(|\mathbf{W}_\tau| = a\right) + f(\mathbf{b})\left(1 - \mathbb{P}^{\mathbf{x}}\left(|\mathbf{W}_\tau| = a\right)\right) \\ &= \mathbb{P}^{\mathbf{x}}\left(|\mathbf{W}_\tau| = a\right)\left(f(\mathbf{a}) - f(\mathbf{b})\right) + f(\mathbf{b}),\end{aligned}$$

where $\mathbf{a}, \mathbf{b}$ are any vectors in $\mathbb{R}^d$ with $|\mathbf{a}| = a$ and $|\mathbf{b}| = b$. Then, by our use of Dynkin's Lemma above, we deduce that

$$\mathbb{P}^{\mathbf{x}}\left(|\mathbf{W}_\tau| = a\right) = \frac{f(\mathbf{b}) - f(\mathbf{x})}{f(\mathbf{b}) - f(\mathbf{a})}.$$

We can then see that

$$\lim_{b\to\infty} \mathbb{P}^{\mathbf{x}}\left(|\mathbf{W}_\tau| = a\right) = \begin{cases} 1 & \text{for } d = 2 \\ \frac{f(\mathbf{x})}{f(\mathbf{a})} & \text{for } d \geq 3, \end{cases}$$

and

$$\lim_{a\to 0} \mathbb{P}^{\mathbf{x}}\left(|\mathbf{W}_\tau| = a\right) = 0.$$

Then,

$$\begin{aligned}\mathbb{P}^{\mathbf{x}}\left(\mathbf{W}_t = \mathbf{0}, \text{ some } t \geq 0\right) &= \mathbb{P}^{\mathbf{x}}\left(\bigcup_{b>|x|}\left\{|\mathbf{W}_t| = 0 \text{ before } |\mathbf{W}_t| = b\right\}\right) \\ &= \lim_{b\to\infty}\lim_{a\to 0}\mathbb{P}^{\mathbf{x}}\left(\left|\mathbf{W}_{\tau_{a,b}}\right| = a\right) \\ &= 0.\end{aligned}$$

4.3: (a) The argument runs exactly as in the previous question, that is, Dynkin's Lemma tells us that

$$f(x) = \mathbb{E}^x\left[f(X_{H_{a,b}})\right] = f(a)\left(1 - \mathbb{P}\left(X_{H_{a,b}} = b\right)\right) + f(b)\mathbb{P}\left(X_{H_{a,b}} = b\right),$$

and the result follows from a rearrangement.

(b) For a Brownian motion we have $\mathcal{A}f(x) = \frac{1}{2}\Delta f(x)$, and so if we take $f(x) = x$, then f is bounded on $[a, b]$ and gives $\mathcal{A}f = 0$. Then from the above we have that

$$\mathbb{P}\left(X_{H_{a,b}} = b\right) = \frac{x - a}{b - a}, \quad \mathbb{P}\left(X_{H_{a,b}} = a\right) = \frac{b - x}{b - a}.$$

(c) For this diffusion process we have $\mathcal{A}f(x) = \mu\frac{\partial f}{\partial x} + \frac{1}{2}\sigma^2\frac{\partial^2 f}{\partial x^2}$, and we want to choose a function f such that $\mathcal{A}f = 0$. A quick integration tells us that $f(x) = \mathrm{e}^{-\frac{2\mu}{\sigma^2}x}$ satisfies this condition (this can be quickly verified by differentiating). Then, arguing as above,

$$\mathbb{P}\left(X_{H_{a,b}} = b\right) = \frac{\mathrm{e}^{-\frac{2\mu}{\sigma^2}x} - \mathrm{e}^{-\frac{2\mu}{\sigma^2}a}}{\mathrm{e}^{-\frac{2\mu}{\sigma^2}b} - \mathrm{e}^{-\frac{2\mu}{\sigma^2}a}}, \quad \mathbb{P}\left(X_{H_{a,b}} = a\right) = \frac{\mathrm{e}^{-\frac{2\mu}{\sigma^2}b} - \mathrm{e}^{-\frac{2\mu}{\sigma^2}x}}{\mathrm{e}^{-\frac{2\mu}{\sigma^2}b} - \mathrm{e}^{-\frac{2\mu}{\sigma^2}a}}.$$

4.4: Since we again have a Brownian motion, we have that

$$\mathcal{A}f(x) = \begin{cases} 0 & \text{for } x \le a \\ 1 & \text{for } x \in (a,b] \\ 0 & \text{for } x \ge b. \end{cases}$$

Then, by Dynkin's Lemma,

$$\mathbb{E}^{x_0}\left[f(W_{H_{0,1}})\right] = f(x_0) + \mathbb{E}^{x_0}\left[\int_0^{H_{0,1}} \mathbf{1}\{W_s \in (a,b]\}\mathrm{d}s\right],$$

but this last term is exactly the expected amount of time the Brownian motion spends in $(a,b]$. Rearranging and using the hitting probabilities calculated earlier, we get

$$\begin{aligned}\mathbb{E}^{x_0}\left[\int_0^{H_{0,1}} \mathbf{1}\{W_s \in (a,b]\}\mathrm{d}s\right] &= \mathbb{E}^{x_0}\left[f(W_{H_{0,1}})\right] - f(x_0) \\ &= x_0 f(1) + (1-x_0)f(0) - f(x_0) \\ &= x_0 f(1) - f(x_0).\end{aligned}$$

4.5: We calculate $\mathcal{A}v$ from the original definition:

$$\begin{aligned}\mathcal{A}v(t,x) &= \lim_{r\to 0}\frac{1}{r}\left(\mathbb{E}^x\left[v(t,X_r)\right] - v(t,x)\right) \\ &= \lim_{r\to 0}\frac{1}{r}\Bigg(\mathbb{E}^x\Bigg[\mathbb{E}^{X_r}\left[f(X_t)\exp\left(-\int_0^t q(X_s)\mathrm{d}s\right)\right] \\ &\qquad\qquad - \mathbb{E}^x\left[f(X_t)\exp\left(-\int_0^t q(X_s)\mathrm{d}s\right)\right]\Bigg]\Bigg) \\ &= \lim_{r\to 0}\frac{1}{r}\mathbb{E}^x\left[f(X_{t+r})\exp\left(-\int_0^t q(X_{s+r})\mathrm{d}s\right) - f(X_t)\exp\left(-\int_0^t q(X_s)\mathrm{d}s\right)\right] \\ &= \lim_{r\to 0}\frac{1}{r}\mathbb{E}^x\left[f(X_{t+r})\exp\left(-\int_0^{t+r} q(X_s)\mathrm{d}s\right) - f(X_t)\exp\left(-\int_0^t q(X_s)\mathrm{d}s\right)\right] \\ &\qquad + \lim_{r\to 0}\frac{1}{r}\mathbb{E}^x\left[f(X_{t+r})\exp\left(-\int_0^{t+r} q(X_s)\mathrm{d}s\right)\left(\exp\left(\int_0^r q(X_s)\mathrm{d}s\right) - 1\right)\right] \\ &= \frac{\partial v}{\partial t}(t,x) + q(x)v(t,x),\end{aligned}$$

since $\left(\exp\left(\int_0^r q(X_s)\mathrm{d}s\right) - 1\right) \to q(X_0)$.

Take w a bounded solution to the same system and fix $(s,x,z) \in \mathbb{R}\times\mathbb{R}^n\times\mathbb{R}^n$. Define $Z_t = z + \int_0^t q(X_s)\mathrm{d}s$ and $H_t = (s-t, X_t^{0,x}, Z_t)$ so that H_t is an Itô diffusion with generator

$$\mathcal{A}_H\phi(s,x,z) = -\frac{\partial\phi}{\partial s} + \mathcal{A}\phi + q(x)\frac{\partial\phi}{\partial z},$$

for appropriate ϕ. If we take $\phi(s,x,z) = \exp(-z)w(s,x)$ then $\mathcal{A}_H\phi = 0$, since w satisfies (1). Then, for $\tau_R = \inf\{t \ge 0 : |H_t| \ge R\}$ we have by Dynkin's Lemma that

$$\mathbb{E}^{s,x,z}\left[\phi(H_{t\wedge\tau_R})\right] = \phi(s,x,z),$$

and so

$$\begin{aligned}
w(s,x) &= \phi(s,x,0) \\
&= \mathbb{E}^{s,x,0}\left[\phi(H_{t\wedge\tau_R})\right] \\
&= \mathbb{E}^x\left[\exp\left(-\int_0^{t\wedge\tau_R} q(X_r)\mathrm{d}r\right) w(s-t\wedge\tau_R, X_{t\wedge\tau_R})\right] \\
&\to \mathbb{E}^x\left[\exp\left(-\int_0^{t} q(X_r)\mathrm{d}r\right) w(s-t, X_t)\right]
\end{aligned}$$

by bounded convergence. In particular, for $s=t$,

$$w(s,x) = \mathbb{E}^x\left[\exp\left(-\int_0^{s} q(X_r)\mathrm{d}r\right) w(0, X_s^{0,x})\right] = v(s,x).$$

4.6: (a) We know by Theorem 6.7 that

$$\begin{aligned}
\mathcal{A}f &= \nabla(\ln h)\cdot\nabla f + \frac{1}{2}\Delta f \\
&= \frac{1}{h}\nabla h\cdot\nabla f + \frac{1}{2}\Delta f \\
&= \frac{1}{2h}\left(2\nabla h\cdot\nabla f + h\Delta f + f\Delta h\right) \\
&= \frac{\Delta(hf)}{2h}.
\end{aligned}$$

If $f=\frac{1}{h}$ then we have $\mathcal{A}f = \frac{\Delta(1)}{2h} = 0$.

(b) Consider the stopping time $\tau_{a,b} = \inf\{t\geq 0 : f(X_t)\notin(a,b)\}$. We plan to take $a\to 0$ and $b\to\infty$. In particular, since $h>0$ in D and since h tends to zero or infinity on the boundary, $\tau_{a,b}<\tau_D$, the exit time for D, for $a,b\in(0,\infty)$, but in the limit $\tau_{a,b}\nearrow\tau_D$. But now consider (for $a<f(x)<b$)

$$\begin{aligned}
f(x) &= \mathbb{E}^x\left[f(X_{\tau_{a,b}})\right] \\
&= a\mathbb{P}^x\left(f(X_{\tau_{a,b}})=a\right) + b\mathbb{P}^x\left(f(X_{\tau_{a,b}})=b\right).
\end{aligned}$$

Take for example $b=a^{-1}$, and let $b\to\infty$. Then $\mathbb{P}^x\left(f(X_{\tau_{b^{-1},b}})=b\right)\to 0$, and hence $\mathbb{P}^x\left(f(X_{\tau_{b^{-1},b}})=b^{-1}\right)\to 1$ as $b\to\infty$.

Noting that $\tau_{b^{-1},b}\nearrow\tau_D$ as $b\to\infty$, it follows that we must have $f(X_{\tau_{b^{-1},b}})\to 0$ as $b\to\infty$. Since $f=h^{-1}$, this is only possible if $h(X_{\tau_{b^{-1},b}})\to\infty$ as $b\to\infty$, or equally, if $X_{\tau_{b^{-1},b}}\to x_0$ as $b\to\infty$. By the continuity of paths, the result follows.

(c) First note that for $\mathbf{x}=(x^1,x^2)\in\partial D$, $\mathbf{x}\neq(0,1)$ we have that $\lim_{\mathbf{z}\to\mathbf{x}} h(\mathbf{z})=0$, but for example by taking the limit towards $(0,1)$ along $x^1=0$ we see that $\lim_{\mathbf{z}\to(0,1)} h(\mathbf{z})=\infty$. Then, by the previous parts, the diffusion X that solves the SDE

$$\mathrm{d}X_t = \nabla(\ln h)\mathrm{d}t + \mathrm{d}W_t$$

will exit D at $(0,1)$ almost surely.